

Module 1 Lesson 6

Exponents and Scientific Notation (5.6)

Objectives

In this lesson, we will learn how to

- use properties of exponents;
- convert from scientific notation to decimal notation;
- convert from decimal notation to scientific notation;
- perform computations using scientific notation;
- solve applied problems using scientific notation.

1. Properties of Exponents

Exponents are used to represent repeated multiplication.

For example,

$$5^3 = 5 \cdot 5 \cdot 5.$$

Product Rule

When multiplying exponential expressions with the same base, add the exponents.

$$b^m \cdot b^n = b^{m+n}.$$

Example:

$$9^6 \cdot 9^{12} = 9^{6+12} = 9^{18}.$$

Power Rule

When an exponential expression is raised to a power, multiply the exponents.

$$(b^m)^n = b^{mn}.$$

Examples:

$$(3^4)^5 = 3^{4 \cdot 5} = 3^{20}$$

$$(5^3)^8 = 5^{3 \cdot 8} = 5^{24}.$$

Quotient Rule

When dividing exponential expressions with the same nonzero base, subtract the exponent in the denominator from the exponent in the numerator.

$$\frac{b^m}{b^n} = b^{m-n}, \quad b \neq 0.$$

2. The Zero Exponent Rule

Zero Exponent Rule

If b is any real number other than 0, then

$$b^0 = 1.$$

Example 1: Use the zero exponent rule to simplify:

$$7^0$$

$$\left(\frac{2}{3}\right)^0$$

$$(-5)^0$$

$$-5^0$$

Solution

$$7^0 = 1$$

$$\left(\frac{2}{3}\right)^0 = 1$$

$$(-5)^0 = 1$$

$$-5^0 = -(5^0) = -1$$

3. The Negative Exponent Rule

Negative Exponent Rule

If b is any real number other than 0, and m is a natural number, then

$$b^{-m} = \frac{1}{b^m}.$$

Also,

$$\frac{1}{b^{-m}} = b^m.$$

A negative exponent does not make the number negative. It tells us to move the factor to the other side of the fraction bar.

Example 2: Use the negative exponent rule to simplify:

$$4^{-2}$$

$$\frac{1}{3^{-4}}$$

$$\left(\frac{2}{5}\right)^{-3}$$

Solution

$$4^{-2} = \frac{1}{4^2} = \frac{1}{16}$$

$$\frac{1}{3^{-4}} = 3^4 = 81$$

$$\left(\frac{2}{5}\right)^{-3} = \left(\frac{5}{2}\right)^3 = \frac{125}{8}$$

4. Powers of Ten

Powers of 10 are especially important in scientific notation.

Positive Powers of Ten

A positive exponent tells how many zeros follow the 1.
For example,

$$10^9 = 1,000,000,000.$$

Negative Powers of Ten

A negative exponent tells how many places there are to the right of the decimal point.
For example,

$$10^{-9} = 0.000000001.$$

5. Scientific Notation

A positive number is written in **scientific notation** when it is expressed in the form

$$a \times 10^n,$$

where

$$1 \leq a < 10$$

and n is an integer.

Scientific Notation

$$a \times 10^n$$

where a is at least 1 but less than 10, and n is an integer.

Examples:

$$4.6 \times 10^6$$

$$2.3 \times 10^{-5}.$$

6. Converting Scientific Notation to Decimal Notation

To convert from scientific notation to decimal notation, use the exponent on 10.

Scientific Notation to Decimal Notation

If n is positive, move the decimal point in a to the right n places.
If n is negative, move the decimal point in a to the left $|n|$ places.

Example 3: Write each number in decimal notation:

$$1.375 \times 10^{10}$$

$$1.1 \times 10^{-4}.$$

Solution

$$1.375 \times 10^{10} = 13,750,000,000$$

$$1.1 \times 10^{-4} = 0.00011$$

7. Converting Decimal Notation to Scientific Notation

To write a number in the form

$$a \times 10^n,$$

we need to find a and n .

Decimal Notation to Scientific Notation

1. Move the decimal point to obtain a number a such that

$$1 \leq a < 10.$$

2. Count the number of places the decimal point moved.
3. If the original number is greater than or equal to 10, then n is positive.
4. If the original number is between 0 and 1, then n is negative.

Example 4: Write each number in scientific notation:

$$4,600,000$$

$$0.000023.$$

Solution

$$4,600,000 = 4.6 \times 10^6$$

$$0.000023 = 2.3 \times 10^{-5}$$

8. Multiplying Numbers in Scientific Notation

To multiply numbers in scientific notation, multiply the decimal parts and use the product rule for powers of 10.

Multiplying in Scientific Notation

$$(a \times 10^n)(b \times 10^m) = (ab) \times 10^{n+m}.$$

Add the exponents on 10, and multiply the other parts separately.

Example 5: Multiply. Write the product in decimal notation:

$$(3.4 \times 10^9)(2 \times 10^{-5}).$$

Solution

$$(3.4 \times 10^9)(2 \times 10^{-5}) = (3.4 \cdot 2) \times 10^{9+(-5)}$$

$$= 6.8 \times 10^4$$

$$= 68,000.$$

9. Dividing Numbers in Scientific Notation

To divide numbers in scientific notation, divide the decimal parts and use the quotient rule for powers of 10.

Dividing in Scientific Notation

$$\frac{a \times 10^n}{b \times 10^m} = \frac{a}{b} \times 10^{n-m}, \quad b \neq 0.$$

Subtract the exponents on 10, and divide the other parts separately.

Example 6: Divide. Write the quotient in decimal notation:

$$\frac{6.4 \times 10^8}{2 \times 10^3}.$$

Solution

$$\begin{aligned}\frac{6.4 \times 10^8}{2 \times 10^3} &= \frac{6.4}{2} \times 10^{8-3} \\ &= 3.2 \times 10^5 \\ &= 320,000.\end{aligned}$$

10. Application: The National Debt

Scientific notation is useful when working with very large or very small numbers.

Example 7: As of April 2021, the national debt was about

\$28.1 trillion

or

28.1×10^{12} dollars.

At that time, the U.S. population was approximately

331,000,000

or

3.31×10^8 .

If the national debt were evenly divided among every individual in the United States, how much would each citizen have to pay?

Solution

The amount each citizen would have to pay is

$$\begin{aligned} & \frac{28.1 \times 10^{12}}{3.31 \times 10^8} \\ &= \frac{28.1}{3.31} \times 10^{12-8} \\ &\approx 8.49 \times 10^4 \\ &= 84,900. \end{aligned}$$

Each citizen would have to pay approximately

\$84,900.

Summary

Key Ideas

- When multiplying powers with the same base, add the exponents:

$$b^m \cdot b^n = b^{m+n}.$$

- When raising a power to a power, multiply the exponents:

$$(b^m)^n = b^{mn}.$$

- When dividing powers with the same nonzero base, subtract the exponents:

$$\frac{b^m}{b^n} = b^{m-n}.$$

- For $b \neq 0$,

$$b^0 = 1.$$

- Negative exponents indicate reciprocals:

$$b^{-m} = \frac{1}{b^m}.$$

- Scientific notation has the form

$$a \times 10^n, \quad 1 \leq a < 10.$$

- Scientific notation is useful for very large or very small numbers.